

Aditya Samalla

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Summary

Engineering leader and Principal ML Engineer with 10+ years building production AI at scale. I founded and lead an LLM + RAG product line built on Claude, am a founding engineer and primary architect of the ML engine behind a flagship cloud-security product, and built the evaluation and red-teaming practice that keeps it safe and reliable. I turn frontier-model capabilities into measured outcomes — fewer false positives, multi-million-dollar savings, and revenue — and lead the teams that ship them.

Core Competencies

- LLMs, RAG, Agentic / MCP, Prompt Engineering
- LLM Evaluation, Red-teaming & AI Safety
- Machine Learning, Deep Learning, MLOps
- Anomaly & Behavior Detection, Graph ML
- Production ML Systems & Data Platforms
- Engineering Management, Roadmaps, Mentoring
- **Languages:** Python, SQL, Scala, Go, R
- **AI/ML:** Claude (Bedrock), Azure OpenAI, LangChain, PyTorch, TensorFlow
- **Data:** Spark, Apache Iceberg/Lakehouse, Snowflake, Postgres, Neo4j, Airflow
- **Infra:** Kubernetes, Docker, AWS/GCP/Azure, Argo CD, Grafana

Work Experience

Fortinet (formerly Lacework)

Software Engineering Manager / Principal Machine Learning Engineer

Promoted Senior → Staff Machine Learning Engineer → current role (3 levels, 2022–2026)

Aug 2022 – Present

Sunnyvale, California

- **Founded and lead the AI Assist product line** — an LLM + RAG system (built on Claude via Amazon Bedrock) that turns complex cloud-security alerts into prioritized incident reports and alert-grounded remediation, exposed through an alert-context API for **MCP / agentic workflows**; set the roadmap at launch and shipped every milestone, driving **30% higher analyst engagement**.
- Built the **LLMOps and AI-safety practice** that gates it: an **LLM-as-judge evaluation harness** run on every prompt change, **adversarial red-teaming (5,000+ prompts, zero successful bypasses)**, production SLOs (p95 latency, error rate), and per-request token/cost observability.
- **Founding engineer and primary architect of the correlation engine** behind Composite Alerts — correlating **400+ detections** across **140+ MITRE ATT&CK techniques** and cloud/host/container/ML-anomaly signals into single high-fidelity incidents, **reducing false positives 40%**.
- Designed an **ephemeral graph-database microservice** within a Kubernetes multi-container framework to surface attack paths in potential breaches — **improving recall 15%, cutting latency 10x, and reducing data-warehouse cost \$1M/year**.
- **Designed and led** the migration of the engine **off Snowflake onto an open Spark + Apache Iceberg lakehouse** — authoring the architecture (a producer/consumer split with single-scan grouped caching) and validating the production cutover — **cutting \$1.5M+ in annualized cloud cost** (–53% warehouse, –57% compute).
- Architected a **semi-supervised classification** approach within a Streamlit labeling application to boost detection efficacy — directly contributing to a **\$1.2M enterprise deal**.
- Built **timeseries anomaly detection** across AWS/GCP/Azure to flag unusual behavior before compromise, **increasing detection coverage 10%**; integrated runtime vulnerable-activity detection into the correlation graph.
- Engineered an automated pipeline for **continuous evaluation** of detection effectiveness — recording precision/recall and ML metrics per detection to surface regressions early and tune efficacy; shipped a customer-facing detection catalog surfacing 400+ detections.
- Manage and mentor a team across detection, GenAI, and platform; own the cross-team product-delivery roadmap, lead org-wide product launches, and set technical direction and code-review standards across the team's services.

Salesforce

Jun 2021 – Aug 2022

Senior Data Scientist

San Francisco, California

- Built and deployed supervised models (AWS SageMaker) forecasting customer subscription renewals for Tableau Online, **reducing churn 1.5%** at product scale.
- Applied topic modeling to autonomously group unstructured community ideas and stakeholder feedback for feature ranking, **increasing enhancement velocity 10%**.
- Designed supervised models using **SHAP** analysis to assess and improve Tableau visualization performance and user experience.
- Built data pipelines and models ingesting public census data for the Tableau Data Equity Hub, which secured a **\$100M MacArthur Foundation grant**.

Hewlett Packard Enterprise

Jun 2017 – Jun 2021

Senior Staff Data Scientist

San Jose, California

- Built a workload **anomaly detection** engine monitoring storage-performance metrics, with performance timelines visualizing anomalies and workload variation.
- Engineered a storage recommendation engine that diagnosed performance issues and recommended optimizations — **automatically resolving 86% of customer cases**.
- Created a virtual-environment recommender that triaged and prioritized exceptions, **resolving 54% of issues** beyond the storage layer; developed an "Impact Score" to predict and quantify severity of affected systems.
- Applied **dimensionality reduction and unsupervised learning** on high-dimensional telemetry to surface natural variability, cluster behavior, and summarize virtual environments into meaningful topologies; built diagnostic signatures through cross-team collaboration with support, performance engineering, and product.
- Developed forecasting and sizing models for capacity/performance and data-driven sales — **driving 15% revenue and 10% sales-efficiency gains**; architected lead-generation integration enabling **\$3M+ in opportunities**.

Nimble Storage

Sep 2016 – May 2017

Data Scientist

San Jose, California

- Built statistical and machine-learning models on historical data to predict storage issues and performance bottlenecks proactively.
- Designed executive data-visualization dashboards (Tableau / R Shiny) and built Flask APIs feeding downstream systems — driving KPI tracking and data-driven decisions across marketing, sales, and pricing with cross-functional partners.
- Ran exploratory data analysis on operational, sales, and support metrics and automated the supporting data pipelines, **improving support satisfaction 12.5%**.

Accomplishments

- **Patent (Allowed, USPTO):** Generative AI-Based Multi-stage Composite Event Processing (Appl. 18/385,069) — co-inventor.
- **Patent (Granted):** US 9,065,952 B2 — robust determination of the significance of performance issues in IT environments. **Patent (Pending):** ML-based detection of compromised AWS credentials.
- **Publication / blog:** "FortiCNAPP AI Assist: Accelerating the Investigation and Remediation of Cloud Threats" (Fortinet Community, 2026); "Performance Analysis of Noise Cancellation in Speech Signals Using LMS, FT-LMS and RLS Algorithms."
- Member, **Beta Gamma Sigma** — International Honor Society (highest scholastic recognition). Presented ML features at HPE Techcon, Discover, and storage conferences.

Education

W. P. Carey School of Business, Arizona State University — M.S. in Analytics, GPA 4.00

2016

Jawaharlal Nehru Technological University — B.Tech, Electronics & Communication Engineering

2011